

MongoDB Operations Documentation with screenshots and explanations.

MongoDB Operations Documentation

Database Used

- **Database Name:** demodb
 - **Collection Name:** people
-

1. Update One Document

Query:

```
db.people.updateOne(  
  
  { user_id: 3 },  
  
  { $set: { status: "failed" } }  
  
)
```

Screenshot:

```
demodb> db.people.updateOne({ user_id: 3 }, { $set: { status: "failed" } })  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
demodb> db.people.find()  
[  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),  
    user_id: 1,  
    age: 25,  
    status: 'passed'  
  },  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),  
    user_id: 2,  
    age: 30,  
    status: 'failed'  
  },  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b4'),  
    user_id: 3,  
    age: 22,  
    status: 'failed'  
  },  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),  
    user_id: 4,  
    age: 35,  
    status: 'failed'  
  },  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b6'),  
    user_id: 5,  
    age: 28,  
    status: 'passed'  
  }  
]
```

Explanation:

This command updates **only the first document** where user_id is 3. It sets the status field to "failed" using the \$set operator.

✅ 2. Update Many Documents

Query:

```
db.people.updateMany(  
  { status: "failed" },  
  { $inc: { age: 2 } }  
)
```

Screenshot:

```
demodb> db.people.updateMany({ status: "failed" }, { $inc: { age: 2 } })  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 3,  
  modifiedCount: 3,  
  upsertedCount: 0  
}  
demodb> db.people.find()  
[  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),  
    user_id: 1,  
    age: 25,  
    status: 'passed'  
  },  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),  
    user_id: 2,  
    age: 33,  
    status: 'failed'  
  },  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b4'),  
    user_id: 3,  
    age: 25,  
    status: 'failed'  
  },  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),  
    user_id: 4,  
    age: 38,  
    status: 'failed'  
  },  
  {  
    _id: ObjectId('687e0e263f06dc6c05eec4b6'),  
    user_id: 5,  
    age: 28,  
    status: 'passed'  
  }  
]
```

Explanation:

This command increases the age of **all documents** having status equal to "failed" by 2 using the \$inc operator.

✖ 3. Delete One Document

🔧 Query:

```
db.people.deleteOne({ age: 28 })
```

🖼 Screenshot:

```
demodb> db.people.deleteOne({ age: 28 })
{ acknowledged: true, deletedCount: 1 }
demodb> db.people.find()
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),
    user_id: 1,
    age: 25,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    user_id: 2,
    age: 33,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b4'),
    user_id: 3,
    age: 25,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),
    user_id: 4,
    age: 38,
    status: 'failed'
  }
]
```

📄 Explanation:

Deletes **the first matching document** where age is 28 from the people collection.

✖ 4. Delete Many Documents

🔧 Query:

```
db.people.deleteMany({ age: 25 })
```

🖼 Screenshot:

```
demodb> db.people.deleteMany({ age : 25 })
{ acknowledged: true, deletedCount: 2 }
demodb> db.people.find()
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    user_id: 2,
    age: 33,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),
    user_id: 4,
    age: 38,
    status: 'failed'
  }
]
```

📄 Explanation:

Deletes **all documents** where age is 25.

🔄 5. Replace One Document

🔧 Query:

```
db.people.replaceOne(
  { user_id: 2 },
  { user_id: 2, age: 31, status: "passed" }
)
```

Screenshot:

```
demodb> db.people.replaceOne(
...   { user_id: 2 },
...   { user_id: 2, age: 31, status: "passed" }
... )
...
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
demodb> db.people.find()
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    user_id: 2,
    age: 31,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),
    user_id: 4,
    age: 38,
    status: 'failed'
  }
]
```

Explanation:

This replaces the **entire document** for user_id: 2 with a new one, removing all existing fields except those provided.

6. Projection Query

Query:

```
db.people.find(
  { status: "passed" },
  { user_id: 1, _id: 0 }
)

db.people.find(
  { status: "passed" },
  { user_id: 1, status: 1, _id: 0 }
)
```

Screenshot:

```
demodb> db.people.find({ status: "passed" }, { user_id: 1, _id: 0 })
[ { user_id: 2 }, { user_id: 6 } ]
demodb> db.people.find({ status: "passed" }, { user_id: 0, _id: 0 })
[ { age: 31, status: 'passed' }, { age: 22, status: 'passed' } ]
demodb> db.people.find({ status: "passed" }, { user_id: 2, _id: 0 })
[ { user_id: 2 }, { user_id: 6 } ]
demodb> db.people.find({ status: "passed" }, { user_id: 0, _id: 0 })
[ { age: 31, status: 'passed' }, { age: 22, status: 'passed' } ]
demodb> db.people.find({ status: "passed" }, { user_id: 0, _id: 1 })
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    age: 31,
    status: 'passed'
  },
  {
    _id: ObjectId('687f574d4a6b6ab7cbeec4a9'),
    age: 22,
    status: 'passed'
  }
]
```

Explanation:

This performs a **projection**, where only `user_id` is included in the output. `_id` is excluded by setting it to 0 and also performs a projection where `_id` is excluded in output by setting it as 0.

7. Using \$in Operator

Query:

```
db.people.find({
  status: { $in: ["passed", "pending"] }
})
```

Screenshot:

```
demodb> db.people.find({ status: { $in: ["passed", "pending"] } })
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    user_id: 2,
    age: 31,
    status: 'passed'
  },
  {
    _id: ObjectId('687f574d4a6b6ab7cbeec4a9'),
    user_id: 6,
    age: 22,
    status: 'passed'
  },
  {
    _id: ObjectId('687f5cbc4a6b6ab7cbeec4ab'),
    user_id: 3,
    name: 'Kiran',
    status: 'pending'
  }
]
```

Explanation:

The \$in operator filters documents where status matches **either "passed" or "pending"**.

Conclusion

This document demonstrates key MongoDB operations:

- Updating (single/multiple)
- Deleting (single/multiple)
- Replacing
- Querying with projections and operators like \$in

All queries were tested on demodb.people collection and verified with screenshots for proof of execution.
